

PONTYAGIN SPECTRAL INDEX OF THE WEIL FORM AND THE CONNES SCALING OPERATOR: A BRIDGE THEOREM FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. Under the assumption that the Riemann ζ -function has exactly m orbits of zeros off the critical line (*Hypothesis D*), the Weil quadratic form Q acquires a Kreĭn negative index $\kappa = 2m$ (established in the companion work). We formulate a *bridge conjecture* identifying $\kappa(Q)$ with the number of complex eigenvalue pairs of the Connes scaling operator H_C , when H_C is viewed as a Q -symmetric operator on a Pontryagin space $(\mathcal{K}, Q)$ built from the pair $(L^2(C_{\mathbb{Q}}), Q)$. The conjecture states

$$\kappa(Q) = \text{neg. ind}(H_C),$$

so that $\text{RH} \iff \text{neg. ind}(H_C) = 0$. The Kreĭn–Langer spectral theorem yields the bound $\text{neg. ind}(H_C) \leq \kappa$ as a free consequence once H_C is Q -symmetric. The reverse inequality—the spectral correspondence between off-line zeros and complex eigenvalues of H_C —requires a Pontryagin-space completion of $(L^2(C_{\mathbb{Q}}), Q)$ together with a distributional extension admitting the characters $x^{b_j + i\gamma_j}$ as generalized eigenvectors. We formulate the verification program **V.1–V.4**, identify **V.2** as the central open problem, and derive the adelic-rigidity direction (*adelic rigidity*: $\text{neg. ind}(H_C) = 0$ from the structure of $C_{\mathbb{Q}}$).

CONTENTS

1.	Introduction	2
2.	The Kreĭn index of the Weil form	2
3.	The Connes scaling operator	3
4.	The Pontryagin space framework	3
5.	The bridge conjecture	4
6.	The verification program V.1–V.4	5
7.	Phase 27: adelic rigidity	5
8.	Open problems and conclusion	6
	References	7

1. INTRODUCTION

The Riemann Hypothesis (RH) states that all non-trivial zeros of $\zeta(s)$ lie on the critical line $\operatorname{Re}(s) = 1/2$. Two independent reformulations of RH motivate this paper.

(Weil–Kreĭn). Weil’s positivity criterion [1] asserts

$$\text{RH} \iff Q(f, f) \geq 0 \text{ for all admissible } f,$$

where $Q(f, g) = \mathcal{W}(f * \tilde{g})$ is the Weil quadratic form. the companion work of this series established that under Hypothesis D (finitely many off-line orbits, see Definition 2.4), the Kreĭn negative index of Q satisfies

$$(1) \quad \kappa(Q) = 2m,$$

where m is the number of Klein four-group orbits of zeros with $\operatorname{Re}(\rho) > 1/2$. RH is equivalent to $\kappa(Q) = 0$.

(Connes–Spectral). Connes [4] showed that the zeros of ζ appear in the spectrum of a scaling operator H_C on the idèle class space $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*$, and argued:

$$(2) \quad \text{RH} \iff H_C \text{ is self-adjoint (spectrum in } i\mathbb{R}).$$

The bridge. The bridge conjecture (Conjecture 5.3) unifies (1) and (2) via a single dictionary:

$$\kappa(Q) = \text{neg. ind}(H_C),$$

where H_C is viewed as a Q -symmetric operator on the Pontryagin space $\mathcal{K}$ constructed from $(L^2(C_{\mathbb{Q}}), Q)$. Corollary 5.4 then gives:

$$\text{RH} \iff \text{neg. ind}(H_C) = 0.$$

Honest status. The bridge theorem is a *conjecture*. The Kreĭn–Langer spectral theorem [2, 3] yields $\text{neg. ind}(H_C) \leq \kappa$ unconditionally (given Q -symmetry). The reverse inequality is the central open problem identified as **V.2**: constructing a distributional extension $\mathcal{K}^+$ that admits the characters $x^{b_j + i\gamma_j}$ as genuine eigenvectors of H_C . Without this construction the bridge remains a precise conjecture, not a theorem.

Organization. Section 2 reviews the Kreĭn index. Section 3 recalls the Connes operator. Section 4 introduces the Pontryagin space framework. Section 5 states the bridge conjecture and its consequences. Section 6 formulates the verification program V.1–V.4. Section 7 outlines the adelic-rigidity step (adelic rigidity). Section 8 collects open problems.

2. THE KREĬN INDEX OF THE WEIL FORM

Let $\mathcal{S} = \mathcal{S}(\mathbb{R}_+^*)$ be the Schwartz space of rapidly decreasing smooth functions on $\mathbb{R}_+^*$. For $f \in \mathcal{S}$, let $\hat{f}(s) = \int_0^\infty f(x)x^{s-1}dx$ be its Mellin transform, and set $\tilde{f}(x) = \overline{\hat{f}(1/x)} \cdot x^{-1}$.

Definition 2.1 (Weil functional and quadratic form). *The Weil functional is*

$$(3) \quad \mathcal{W}(f) = \sum_{\rho} \hat{f}(\rho) - \hat{f}(0) \log \pi - \hat{f}(1) \log \pi + \sum_{p,n} \frac{\log p}{p^{n/2}} [f(p^n) + f(p^{-n})] + f(1) \log \pi,$$

where the first sum runs over all non-trivial zeros of $\zeta(s)$, and $\sum_{p,n}$ runs over primes p and integers $n \geq 1$. The Weil quadratic form is

$$Q(f, g) := \mathcal{W}(f * \tilde{g}), \quad f, g \in \mathcal{S}.$$

Theorem 2.2 (Weil [1]). *$Q(f, f) \geq 0$ for all $f \in \mathcal{S}$ if and only if the Riemann Hypothesis holds.*

Definition 2.3 (Kreĭn negative index). *The Kreĭn negative index of Q is*

$$\kappa(Q) := \sup \{k \in \mathbb{N} : \exists f_1, \dots, f_k \in \mathcal{S} \text{ with } \det(Q(f_i, f_j))_{1 \leq i, j \leq k} < 0\}.$$

By Theorem 2.2: $\text{RH} \iff \kappa(Q) = 0$.

Definition 2.4 (Hypothesis D). *We say Hypothesis D holds with parameter $m \geq 1$ if the set of non-trivial zeros ρ of ζ with $\text{Re}(\rho) \neq 1/2$ consists of exactly m complete orbits under the Klein four-group $V = \{\text{id}, \sigma, \iota, j\}$ acting by $\rho \mapsto \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$. Each orbit has four elements: $\rho_j, \bar{\rho}_j, 1 - \rho_j, 1 - \bar{\rho}_j$, with $\rho_j = (1/2 + b_j) + i\gamma_j$, $b_j \in (0, 1/2)$, $\gamma_j > 0$.*

Theorem 2.5 (P32, P33). *Under Hypothesis D with m orbits, $\kappa(Q) = 2m$.*

This was established in the companion work via the Kreĭn–Langer factorization $\xi = \xi_0 \cdot P_{4m}$, where P_{4m} is a polynomial of degree $4m$ encoding the off-line zeros and ξ_0 is in the Laguerre–Pólya class. The polynomial P_{4m} is strictly positive on the critical line, and the factorization assigns each pair of off-line zeros to one negative direction of Q .

3. THE CONNES SCALING OPERATOR

Let $\mathbb{A}_{\mathbb{Q}}$ be the ring of adèles of $\mathbb{Q}$ and $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^*/\mathbb{Q}^*$ the idèle class group. Let $H = L^2(C_{\mathbb{Q}}, d\mu)$ where $d\mu$ is Haar measure.

The group $\mathbb{R}_+^*$ acts on $C_{\mathbb{Q}}$ by scaling: $(U_{\lambda}f)(x) = f(\lambda^{-1}x)$, $\lambda \in \mathbb{R}_+^*$.

Definition 3.1 (Connes scaling operator). *The Connes scaling operator is the infinitesimal generator of (U_{λ}) :*

$$H_C := \frac{1}{i} \frac{d}{d\lambda} \Big|_{\lambda=1} U_{\lambda},$$

acting formally as $(H_C f)(x) = -ix \partial_x f(x)$ (the Euler–Mellin operator on $C_{\mathbb{Q}}$).

Remark 3.2. *Connes [4] established a trace formula expressing the Weil explicit formula as a distributional trace of the scaling action. In Connes’ spectral parametrization, zeros of ζ on the critical line $\text{Re}(s) = 1/2$ appear at imaginary points $\rho - 1/2 = i\gamma$ of the spectrum; RH corresponds to the spectrum of H_C lying entirely in $i\mathbb{R}$.*

The formal action of H_C on a character $f_s(x) = x^{s-1/2}$ of $\mathbb{R}_+^*$ gives:

$$H_C f_s = -ix \partial_x (x^{s-1/2}) = -i(s - 1/2) f_s,$$

so the formal eigenvalue of f_s is $-i(s - 1/2)$. For $s = \rho_j = (1/2 + b_j) + i\gamma_j$:

$$\text{formal eigenvalue} = -i(b_j + i\gamma_j) = \gamma_j - ib_j,$$

which has $\text{Im}(\gamma_j - ib_j) = -b_j \neq 0$. This non-real eigenvalue reflects the off-line displacement of ρ_j .

4. THE PONTRYAGIN SPACE FRAMEWORK

The bridge replaces the standard $L^2(C_{\mathbb{Q}})$ inner product by the Weil form Q , turning (H, Q) into an indefinite inner product space of Pontryagin type.

Proposition 4.1 (Weil form as indefinite inner product). *The bilinear map*

$$\langle f, g \rangle_Q := Q(f, g) = \mathcal{W}(f * \tilde{g})$$

is a Hermitian form on $\mathcal{S}$. Under Hypothesis D:

- $\langle f, f \rangle_Q < 0$ for certain $f \in \mathcal{S}$ (by Theorem 2.5);
- the negative index of $\langle \cdot, \cdot \rangle_Q$ on any finite-dimensional subspace is bounded by $\kappa(Q) = 2m$.

A Pontryagin space completion of (H, Q) requires more than finite negative index.

Definition 4.2 (Kreĭn–Connes space). *If the Hermitian form Q on $H = L^2(C_{\mathbb{Q}})$ satisfies all three conditions:*

- (a) $\ker Q := \{f \in H : Q(f, g) = 0 \ \forall g\}$ is finite-dimensional or trivial;
- (b) the quotient $H/\ker Q$ is complete in the semi-norm $|Q(\cdot, \cdot)|^{1/2}$;
- (c) there exists a fundamental decomposition $H = H_+ \oplus H_-$ (orthogonal with respect to Q) with $Q|_{H_+} > 0$, $-Q|_{H_-} > 0$, and $\dim H_- = \kappa(Q)$;

then the resulting Pontryagin space of negative index κ is denoted $\mathcal{K} = (\mathcal{K}, Q)$.

Remark 4.3. *The existence of $\mathcal{K}$ is not yet established. Conditions (a)–(c) form the first part of the verification program V.1–V.4; specifically, verifying that $\ker Q$ is finite-dimensional requires controlling the Weil functional on the full infinite-dimensional space H , which goes beyond the finite-rank results of P32–P33.*

The next objective is the Q -invariance of the scaling action, which would make H_C automatically Q -symmetric.

Objective 4.4 (Q -invariance of scaling — **V.1**). *Prove that for all $f, g \in \mathcal{S}$ and all $\lambda \in \mathbb{R}_+^*$:*

$$Q(U_\lambda f, U_\lambda g) = Q(f, g).$$

If this holds, then (by differentiating at $\lambda = 1$) H_C is Q -symmetric: $Q(H_C f, g) = Q(f, H_C g)$ for all f, g in the domain of H_C .

Remark 4.5 (Why **V.1** is non-trivial). *The Weil functional (3) contains three types of terms. The prime term transforms as:*

$$(f * \tilde{g})(p^n) \mapsto (U_\lambda f * \widetilde{U_\lambda g})(p^n) = (f * \tilde{g})(\lambda^{-1} p^n)$$

under U_λ , shifting the evaluation from p^n to $\lambda^{-1} p^n$. The global invariance requires either that each term type is separately invariant, or that the shifts cancel in the full sum. This is related to the Tate–Weil adelic transformation formula and constitutes an independent result.

5. THE BRIDGE CONJECTURE

We recall the key result from the theory of operators on Pontryagin spaces.

Theorem 5.1 (Kreĭn–Langer [2, 3]). *Let T be a Q -self-adjoint operator on a Pontryagin space $(\mathcal{K}, Q)$ of negative index $\kappa < \infty$. Then:*

- (i) T has **at most** κ pairs of non-real eigenvalues $\{\mu_j, \bar{\mu}_j\}$ with $\text{Im}(\mu_j) \neq 0$.
- (ii) The essential spectrum $\sigma_{\text{ess}}(T) \subseteq \mathbb{R}$.

Remark 5.2 (At most vs. exactly). *Theorem 5.1(i) gives $\leq \kappa$, not $= \kappa$. The equality requires that the space $\mathcal{K}$ is minimal for T (generated by eigenvectors), or that the construction of $\mathcal{K}$ guarantees that every negative direction of Q produces a genuine complex eigenvalue. This is the content of **V.2** below.*

Conjecture 5.3 (Bridge Theorem). *Assume Hypothesis D with $\kappa = 2m$. Assume further that $\mathcal{K}$ exists (Definition 4.2) and that H_C is Q -self-adjoint in $\mathcal{K}$ (Objective 4.4). Then:*

- (i) *The essential spectrum of H_C in $\mathcal{K}$ is contained in $i\mathbb{R}$, corresponding to the on-line zeros $\{i\gamma\}$.*
- (ii) *H_C has exactly $\kappa = 2m$ complex eigenvalue pairs in $\mathcal{K}$, given by*

$$\{\gamma_j - ib_j, \gamma_j + ib_j\}_{j=1}^m,$$

corresponding to the m off-line orbits $\rho_j = (1/2 + b_j) + i\gamma_j$.

- (iii) *Each complex eigenvalue has multiplicity 1 if the corresponding zero of ζ is simple.*
- (iv) *$\text{neg. ind}_Q(H_C) = \kappa(Q) = 2m$.*

Corollary 5.4 (RH via the spectral index of H_C). *Under Conjecture 5.3:*

$$\text{RH} \iff \kappa(Q) = 0 \iff \text{neg. ind}(H_C) = 0 \iff H_C \text{ is self-adjoint in } \mathcal{K}.$$

Proof. Weil's criterion (Theorem 2.2): $\text{RH} \iff \kappa(Q) = 0$. Theorem 2.5: $\kappa(Q) = 2m$ under Hypothesis D. Conjecture 5.3(iv): $\text{neg. ind}(H_C) = \kappa(Q)$. If $\kappa = 0$: $\mathcal{K}$ degenerates to a Hilbert space, H_C has no complex eigenvalues, and self-adjointness follows from Q -symmetry (Objective 4.4). $\square$

Remark 5.5 (What the bridge achieves). *Before the bridge:* $\text{RH} \iff \kappa(Q) = 0$ and $\text{RH} \iff H_C$ self-adjoint are two reformulations with no direct connection. *After the bridge:* both are equivalent to $\text{neg. ind}(H_C) = 0$, with an explicit dictionary:

- negative directions of $Q \longleftrightarrow$ complex eigenvalues of H_C ,
- off-line zero orbits $\longleftrightarrow$ non-unitary representations of $\mathbb{R}_+^*$ in the scaling spectrum.

This allows tools from adelic arithmetic (Connes program) to act on the Kreĭn index.

6. THE VERIFICATION PROGRAM V.1–V.4

V.1 — Scaling invariance. Prove Objective 4.4: $Q(U_\lambda f, U_\lambda g) = Q(f, g)$ for all $f, g \in \mathcal{S}$, $\lambda \in \mathbb{R}_+^*$.

V.2 — Spectral correspondence (primary objective). This is the central open problem. The candidate eigenvectors are $f_j(x) = x^{b_j + i\gamma_j}$, which satisfy $H_C f_j = (\gamma_j - ib_j)f_j$ formally. However:

$$\|f_j\|_{L^2(C_{\mathbb{Q}})}^2 = \int_0^\infty x^{2b_j-1} dx = +\infty.$$

The character f_j is *not* in $L^2(C_{\mathbb{Q}})$, and $Q(f_j, f_j)$ is not defined in the classical sense.

The distributional construction. One must construct a rigged Hilbert space (Gel'fand triple) $\Phi \subset L^2(C_{\mathbb{Q}}) \subset \Phi'$ such that:

- the characters $f_j = x^{b_j + i\gamma_j}$ lie in Φ' ;
- the regularization $Q_{\text{reg}}(f_j, f_j) = \mathcal{W}_{\text{reg}}(f_j * \tilde{f}_j)$ is finite (possibly zero, so f_j is an isotropic vector of Q);
- the action $H_C : \Phi' \rightarrow \Phi'$ is well-defined, with $H_C f_j = (\gamma_j - ib_j)f_j$.

If this construction succeeds, $\text{neg. ind}(H_C) \geq m$ (each off-line orbit contributes at least one complex eigenvalue pair), and combined with Theorem 5.1(i) one obtains $\text{neg. ind}(H_C) = 2m = \kappa$.

If the construction fails (e.g., $Q_{\text{reg}}(f_j, f_j)$ diverges for all reasonable regularizations), the bridge in the form of Conjecture 5.3 is false as stated, and requires reformulation via localized Weil forms (see Section 7, Strategy B).

V.3 — Index identification. Given **V.2**, apply Theorem 5.1 to H_C in $\mathcal{K}$: $\text{neg. ind}(H_C) \leq \kappa$ (free) and $\text{neg. ind}(H_C) \geq \kappa$ (from **V.2**). Hence $\text{neg. ind}(H_C) = \kappa$.

V.4 — Orbit count. Under Hypothesis D with m orbits, verify that the construction yields exactly m eigenvalue pairs (not $2m$). Each orbit $\{\rho_j, \bar{\rho}_j, 1 - \rho_j, 1 - \bar{\rho}_j\}$ contributes two characters $x^{b_j \pm i\gamma_j}$, but the functional equation symmetry $s \leftrightarrow 1 - s$ of ξ identifies the pairs $(\rho_j, 1 - \bar{\rho}_j)$ and $(\bar{\rho}_j, 1 - \rho_j)$, reducing four characters to two independent eigenvalues: $\gamma_j - ib_j$ and $\gamma_j + ib_j$.

7. PHASE 27: ADELIC RIGIDITY

Assuming Conjecture 5.3, RH reduces to the single spectral question:

$$(4) \quad \text{neg. ind}(H_C) = 0.$$

This is no longer a statement about zeros of ζ ; it is a statement about an operator on $C_{\mathbb{Q}}$. the adelic-rigidity step proposes to establish (4) via the adelic structure of $C_{\mathbb{Q}}$.

27-A (Classification of negative directions). By **V.2**, each negative direction of Q in $\mathcal{K}$ corresponds to an isolated complex eigenvalue of H_C , i.e., to a non-unitary character $|x|^{b_j + i\gamma_j}$ of $\mathbb{R}_+^*$ ($b_j > 0$) in the scaling spectrum.

27-B (Adelic interpretation). In the adèlic framework, the character $|x|^{b_j+i\gamma_j}$ corresponds to a Grössencharacter of $C_{\mathbb{Q}}$ with non-trivial modular growth ($|x|^{b_j}$ factor, $b_j > 0$). Such a character is *non-unitary*: it lies outside the unitary dual $\widehat{C_{\mathbb{Q}}}$ measured by Haar measure.

27-C (Global constraint). The local-global product structure

$$C_{\mathbb{Q}} \cong \prod_p \mathbb{Z}_p^* \times \mathbb{R}_+^*$$

induces a factorization of the Weil form: $Q = Q_{\infty} \times \prod_p Q_p$ (if it exists). Each local factor Q_p encodes the prime- p term of $\mathcal{W}$. The the adelic-rigidity step hypothesis:

$$Q_p \text{ is positive-definite for all } p \implies Q \text{ is positive-definite globally.}$$

This is an *adelic Hasse–Minkowski principle* for the Weil functional form.

27-D (Conclusion). If **27-C** holds and each Q_p is positive-definite (which would follow from the local Riemann hypothesis for $\mathbb{F}_p$, proven by Weil), then $Q \geq 0$, hence $\kappa(Q) = 0$, hence $\text{neg.ind}(H_C) = 0$, hence RH.

Remark 7.1 (Relation to Wall W4-RSRP). *the adelic-rigidity step is the most concrete formulation to date of the fundamental obstacle identified in the companion work (Wall W4-RSRP) and exhaustively surveyed in the companion work (the prior audit): the inability to propagate arithmetic constraints from $\text{Re}(s) > 1$ into the critical strip. the adelic-rigidity step reformulates this as: “does the local positivity of each Q_p imply global positivity of Q ?” This is a local-to-global theorem for an indefinite functional form over $\mathbb{Q}$, analogous to the Hasse–Minkowski theorem (which applies to classical quadratic forms but not to infinite-dimensional functional forms). The infinite-dimensional nature of Q is the key obstacle.*

Remark 7.2 (Comparison with previous phases). *Phases 21–24 exhausted all spectral approaches to a lower bound on $b_j = \text{Re}(\rho_j) - 1/2$ from within the Kreĭn arc, with uniformly negative results. the prior audit audited fifteen classical mechanisms and named Wall W4-RSRP in six equivalent formulations. the present construction reformulates W4-RSRP as a spectral question about H_C : are there non-unitary characters in the scaling spectrum? the adelic-rigidity step proposes to answer this using class field theory and the local-global structure of $C_{\mathbb{Q}}$, offering a genuinely new tool that was not available in Phases 21–25.*

8. OPEN PROBLEMS AND CONCLUSION

The bridge conjecture raises the following mathematically precise questions.

- (1) **Existence of $\mathcal{K}$.** Does the Weil form Q satisfy conditions (a)–(c) of Definition 4.2? In particular, is $\ker Q$ finite-dimensional?
- (2) **V.1 — Scaling invariance.** Is $Q(U_{\lambda}f, U_{\lambda}g) = Q(f, g)$ for all $f, g \in \mathcal{S}$? (Related to the Tate–Weil adelic transformation formula.)
- (3) **V.2 — Spectral correspondence.** Does a rigged Hilbert space completion $\mathcal{K}^+$ exist that admits $f_j = x^{b_j+i\gamma_j}$ as generalized eigenvectors of H_C with $Q_{\text{reg}}(f_j, f_j)$ finite?
- (4) **Adelic Hasse–Minkowski.** Does $Q = \prod_p Q_p$ factorize adelically, and does local positivity of each Q_p imply global positivity of Q ?

Each question is independent of the Riemann Hypothesis. Questions 1–3 constitute the the present construction verification program; Question 4 is the the adelic-rigidity step target.

Conclusion. We have formulated the bridge conjecture $\kappa(Q) = \text{neg.ind}(H_C)$ connecting the Kreĭn index of the Weil form to the spectral index of the Connes scaling operator. Under this conjecture, $\text{RH} \iff \text{neg.ind}(H_C) = 0$, reducing the Riemann Hypothesis to a single spectral

question about an explicit operator on the idèle class space. the adelic-rigidity step proposes to address this via adelic rigidity, asking whether the local Euler factors force global spectral positivity of the Weil form. This is a new tool in the program: a local-to-global principle for an infinite-dimensional Hermitian form over $\mathbb{Q}$.

The paper does not prove RH. It provides a precise new route: a dictionary between three reformulations (Weil positivity, Kreĭn index, Connes spectral index) and a concrete first target for the adelic-rigidity step that, if achieved, would imply RH via a chain of unconditional results.

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